

Evelyn Yoo

Date _____ Period _____

Simplify.

1) $\sqrt{54n}$

2) $\sqrt{50a}$

3) $\sqrt{245x^3}$

4) $\sqrt{196n}$

5) $\sqrt{98n}$

6) $2\sqrt{24} - 3\sqrt{6} - \sqrt{54}$

$$7) 2\sqrt{12} - \sqrt{27} - 3\sqrt{20}$$

$$8) -\sqrt{54} + 2\sqrt{5} - 3\sqrt{24}$$

$$9) -\sqrt{45} - 2\sqrt{45} + 2\sqrt{5}$$

$$10) -2\sqrt{8} - \sqrt{6} - 3\sqrt{2}$$

$$11) (2\sqrt{2} - 5)(\sqrt{2} + 3)$$

$$12) (\sqrt{5} + 1)(\sqrt{5} - 1)$$

$$13) (-3\sqrt{3} - 5\sqrt{2})(-3\sqrt{3} + \sqrt{2})$$

$$14) (4\sqrt{2} + 3)(-2\sqrt{2} + 5)$$

$$15) (2\sqrt{2} + \sqrt{3})(\sqrt{2} + \sqrt{3})$$

$$16) \frac{2}{3 + \sqrt{2}}$$

$$17) \frac{3}{2 - \sqrt{2}}$$

$$18) \frac{\sqrt{2}}{4 - 3\sqrt{3}}$$

$$19) \frac{3}{3 + \sqrt{2}}$$

$$20) -\frac{3}{-3 + 3\sqrt{3}}$$

$$21) \frac{\sqrt{3}-2}{2\sqrt{2}-3\sqrt{3}}$$

$$22) \frac{3-2\sqrt{2}}{3-3\sqrt{2}}$$

$$23) \frac{3-\sqrt{2}}{-2+\sqrt{3}}$$

$$24) \frac{2\sqrt{2}-3\sqrt{3}}{2-2\sqrt{2}}$$

$$25) \frac{\sqrt{3}-2\sqrt{2}}{3+2\sqrt{3}}$$

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Simplify.

1) $\sqrt{54n}$

$3\sqrt{6n}$

2) $\sqrt{50a}$

$5\sqrt{2a}$

3) $\sqrt{245x^3}$

$7x\sqrt{5x}$

4) $\sqrt{196n}$

$14\sqrt{n}$

5) $\sqrt{98n}$

$7\sqrt{2n}$

6) $2\sqrt{24} - 3\sqrt{6} - \sqrt{54}$

$-2\sqrt{6}$

$$7) \ 2\sqrt{12} - \sqrt{27} - 3\sqrt{20}$$

$$\sqrt{3} - 6\sqrt{5}$$

$$8) \ -\sqrt{54} + 2\sqrt{5} - 3\sqrt{24}$$

$$-9\sqrt{6} + 2\sqrt{5}$$

$$9) \ -\sqrt{45} - 2\sqrt{45} + 2\sqrt{5}$$

$$-7\sqrt{5}$$

$$10) \ -2\sqrt{8} - \sqrt{6} - 3\sqrt{2}$$

$$-7\sqrt{2} - \sqrt{6}$$

$$11) \ (2\sqrt{2} - 5)(\sqrt{2} + 3)$$

$$-11 + \sqrt{2}$$

$$12) \ (\sqrt{5} + 1)(\sqrt{5} - 1)$$

$$4$$

$$13) \ (-3\sqrt{3} - 5\sqrt{2})(-3\sqrt{3} + \sqrt{2})$$

$$17 + 12\sqrt{6}$$

$$14) \ (4\sqrt{2} + 3)(-2\sqrt{2} + 5)$$

$$-1 + 14\sqrt{2}$$

$$15) (2\sqrt{2} + \sqrt{3})(\sqrt{2} + \sqrt{3})$$

$$7 + 3\sqrt{6}$$

$$16) \frac{2}{3 + \sqrt{2}}$$

$$\frac{6 - 2\sqrt{2}}{7}$$

$$17) \frac{3}{2 - \sqrt{2}}$$

$$\frac{6 + 3\sqrt{2}}{2}$$

$$18) \frac{\sqrt{2}}{4 - 3\sqrt{3}}$$

$$\frac{-4\sqrt{2} - 3\sqrt{6}}{11}$$

$$19) \frac{3}{3 + \sqrt{2}}$$

$$\frac{9 - 3\sqrt{2}}{7}$$

$$20) -\frac{3}{-3 + 3\sqrt{3}}$$

$$\frac{-1 - \sqrt{3}}{2}$$

$$21) \frac{\sqrt{3} - 2}{2\sqrt{2} - 3\sqrt{3}}$$

$$\frac{-2\sqrt{6} - 9 + 4\sqrt{2} + 6\sqrt{3}}{19}$$

$$22) \frac{3 - 2\sqrt{2}}{3 - 3\sqrt{2}}$$

$$\frac{1 - \sqrt{2}}{3}$$

$$23) \frac{3 - \sqrt{2}}{-2 + \sqrt{3}}$$

$$-6 - 3\sqrt{3} + 2\sqrt{2} + \sqrt{6}$$

$$24) \frac{2\sqrt{2} - 3\sqrt{3}}{2 - 2\sqrt{2}}$$

$$\frac{-2\sqrt{2} - 4 + 3\sqrt{3} + 3\sqrt{6}}{2}$$

$$25) \frac{\sqrt{3} - 2\sqrt{2}}{3 + 2\sqrt{3}}$$

$$\frac{-3\sqrt{3} + 6 + 6\sqrt{2} - 4\sqrt{6}}{3}$$